

Clement Marie

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Heavy background in maths and computer science - Seeking Research Engineer positions from October 2026

Education

École Normale Supérieure Paris-Saclay (*Shanghai Ranking #12*) *Sept 2025 – Aug 2026*
Master M2 – Mathematics, Vision, Learning (MVA) Paris, France
Relevant coursework: Reinforcement Learning; Probabilistic Graphical Models & Deep Generative Models; Geometric Data Analysis; Optimal Transport; Bayesian Machine Learning; Graphs in Machine Learning

Université Paris-Saclay (*Shanghai Ranking #12*) *2024 – 2025*
Master M1 – Artificial Intelligence *Grade: 15.3/20* Paris, France
Relevant coursework: Applied Statistics; High Performance Computing; Optimization; Machine Learning Algorithms; Hands-on NLP

Sorbonne Université (*Shanghai Ranking #41*) *2021 – 2024*
BSc. Double Major in Mathematics and Computer Science *Grade: 15.5/20* Paris, France
Relevant coursework: Topology; Differential Calculus; Measure Theory; Probability Theory; Linear Algebra; Algorithms; Data Structures

Research Experience

CEA LIST (*in partnership with Valeo*) – **Lab of Vision and Learning for Scene Analysis (LVA)** Saclay, France
Research Engineer Intern, supervised by PhD *Apr–Oct 2026 (ongoing)*

- Implementing cutting-edge research methods into efficient, production-ready solutions for 3D scene understanding and autonomous driving, with focus on sensor fusion (camera-radar).
- Developing scalable systems for experiments and benchmarking on real-world autonomous driving datasets (3D detection, 3D semantic segmentation).
- Designing physics-informed deep learning architectures for multi-sensor fusion, combining semantic cues from cameras with range and velocity measurements from radars.

Huawei Research Center Munich, Germany
Research Engineer Intern, supervised by PhD *May–Aug 2025*

- Extended Meta AI's OrienterNet with novel design choices for image-to-map localization.
- Designed multimodal architectures combining images and vector maps using Fourier encodings and Graph Neural Networks (GNNs).
- Focused on architectural innovation to tackle spatial reasoning and multimodal fusion beyond localization.
- Deployed reproducible training and inference environments using Docker for scalable experimentation.

Sorbonne Center of Artificial Intelligence (SCAI) Paris, France
Research and Developer Intern *May–Jul 2024*

- Built transformer-based tool for large-scale text comparison with efficient complexity.
- Deployed interactive web demo using JavaScript and Python.

Awards

Huawei TechArena – 2nd Prize Munich, Germany, 2025
HRTF Spatial Audio – \$3k award

- Presented project at award ceremony before judges.
- Trained models on joint image and HRTF spatial audio data.

Skills

Programming: Python (Fluent – PyTorch, MMCV)

ML/AI: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, pandas, SciPy, FastAPI

Languages: French (Native), English (Fluent)